Music Genre Recognition

Final project report for Selected Topics in Music and Acoustic Engineering course (A.Y. 2025-2026).
Group Members: Eleonora Berra, Federico Giacopuzzi, Tekla Gizella Kalmár, Antonio Treviglio

1. Project motivation

The challenge of automated music genre classification is a cornerstone of modern Music Information Retrieval (MIR). As digital archives grow, the necessity for robust, scalable systems to categorize and organize audio content becomes increasingly critical for recommendation engines and archival management.

1.1. Task

This project focuses on the design and evaluation of a machine learning system for music genre classification. The dataset to be employed is the FMA-small collection, which contains 8000 audio tracks of 30 seconds each, equally distributed across 8 different music genres.
The main objective is to develop a classification pipeline capable of predicting the genre label given an input music recording. Students are encouraged to critically investigate which types of audio features are most informative for genre discrimination. In particular, special attention should be given to representations that capture rhythm (e.g., tempo, beat patterns), harmony (e.g., chroma features), and timbre (e.g., MFCCs, spectral contrast).
Key aspects to consider include:

- Comparing different feature extraction strategies that emphasize rhythmic, harmonic, or timbral content.
- Exploring the combination of multiple feature types to improve classification performance.
- Designing and training classification models, such as deep neural networks or ensemble methods, adapted to the nature of the extracted features.

The final deliverable should be a complete and reproducible system, including a detailed evaluation procedure based on appropriate metrics for multi-class classification tasks.

1.2. Problem Statement and Relevance

While genre is a subjective label, it is grounded in objective acoustic properties. The core problem lies in the difficulty of capturing the high-level musical structure from low-level audio data. Traditional approaches rely on expert-derived statistical descriptors (e.g., timbral, harmonic, and rhythmic features), which often struggle to account for the intra-genre variance and the complex temporal evolution of music. Consequently, the field is currently witnessing a transition from manual feature engineering toward deep learning architectures capable of autonomous pattern recognition.

1.3. Research Objective and Questions

The primary objective of this project is to design and evaluate an effective classification framework that optimizes performance by balancing feature representation and model selection. We seek to move beyond the limitations of classical architectures and surpass robust performance baselines through a more holistic approach to audio analysis.

To achieve this, the work is guided by the following research inquiries:

- **Feature Representation Efficacy:** How do manual statistical descriptors (rhythmic, harmonic, and timbral) compare to autonomous hierarchical spectral representations in their ability to capture genre-specific information?
- **Model Selection and Robustness:** Does the transition from classical ensemble methods (RF, SVM, MLP) to deep convolutional architectures provide the necessary representational capacity to maximize classification accuracy?
- **Generalization Performance:** Can an end-to-end learning framework improve model robustness and mitigate the overfitting observed in baseline systems, thereby ensuring the system performs reliably on unseen audio data?

By addressing these questions, our motivation is to optimize the audio classification pipeline through rigorous feature analysis, ensuring both superior predictive performance and the system-wide robustness required for complex Music Information Retrieval (MIR) tasks.

2. Methodology Overview

2.1. Baseline Approach

The baseline approach is rooted in **traditional feature engineering**, where the classification task is decomposed into two sequential steps: the extraction of expert-derived acoustic descriptors and the subsequent training of classical supervised learning models.

We compute the statistical moments (mean and variance) of key acoustic descriptors to characterize the audio content:

- **Timbral Descriptors:** MFCCs for spectral shape, Spectral Centroid for brightness, and Zero Crossing Rate for noisiness and percussiveness.
- **Harmonic & Rhythmic Descriptors:** Chroma Features to represent tonal distributions and Tempograms to quantify rhythmic evolution.

These descriptors are aggregated into a standardized feature vector (`X_train_ml_scaled`), which is then used to train three distinct classifiers: **Random Forest (RF)**, **Support Vector Machine (SVM)**, and **Multi-Layer Perceptron (MLP)**. This ensemble and kernel-based baseline establishes a rigorous performance benchmark, allowing us to quantify the classification limits imposed by static, human-defined representations.

2.2. State-of-the-Art (SOTA) Method

The proposed SOTA method utilizes a **2D-Convolutional Neural Network (CNN)**, shifting the paradigm from manual feature engineering to **end-to-end autonomous feature learning**.

Unlike the baseline models, which rely on the information loss inherent in statistical aggregation, the CNN operates directly on 2D log-scaled Mel-Spectrograms. This approach preserves the spatial and temporal correlations within the time-frequency domain. The architecture processes the input as a 4D tensor (`batch_size`, `n_mels`, `time_steps`, 1), enabling the network to autonomously identify hierarchical spectral patterns, progressing from low-level texture identification to high-level rhythmic and harmonic structural analysis. This transition effectively bypasses the feature-representation bottleneck, providing a robust framework for complex multi-class genre discrimination.

3. Experimental setup

3.1. Data Acquisition and Pre-processing Pipeline

To ensure data integrity, the ingestion pipeline is executed in two critical phases:

3.1.1. Environment Initialization & Metadata Loading

We establish a standardized foundation by defining global acoustic constants. Metadata is loaded via pandas with MultiIndex support (header=[0, 1]) to correctly parse the FMA hierarchical structure. We perform dataset filtering on the 'set', 'subset' column to isolate the FMA-small collection. Utilizing track_id as the primary index (index_col=0) is a fundamental design choice, enabling efficient relational joins between metadata and audio file paths.

3.1.2. Annotation Alignment & Dataset Cleaning

This phase synchronizes raw audio with corresponding categorical labels. We recursively index physical audio files using pathlib.Path.rglob and perform an inner join between these files and the filtered metadata. To ensure categorical integrity, we flatten the MultiIndex structure and perform a cleanup operation to discard entries lacking valid genre labels. This relational approach effectively eliminates "orphan" files, resulting in a verified, high-quality dataset of *N=8,000* samples.

3.2. Dataset Description

The system is trained and evaluated using the **FMA-small** (Free Music Archive) collection. This dataset consists of 8,000 high-quality audio tracks, each 30 seconds in duration, providing a balanced distribution across 8 distinct musical genres.

The selection of this dataset is strategic, as it offers a standardized benchmark for Music Information Retrieval (MIR) tasks. Each track is annotated with a specific genre label, allowing for supervised learning of the mapping between acoustic characteristics and stylistic categories. The dataset covers a wide sonic spectrum, including Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, and Rock.

To ensure temporal consistency, audio files are resampled to 22050Hz and normalized using Root Mean Square (RMS) scaling to maintain a consistent dynamic range.

3.3. Data Partitioning and Reproducibility

To establish a robust evaluation framework, the FMA-small dataset is partitioned into distinct subsets, ensuring unbiased model training, hyperparameter tuning, and final performance assessment.

3.3.1. Stratified Partitioning

We utilize a stratified splitting strategy (`stratify=df['genre']`), which preserves the original class distribution across all subsets. This is critical for maintaining representativeness, particularly given the multi-class nature of the dataset.

3.3.2. Experimental Setup

The workflow implements a three-way split to ensure rigorous validation:

- **Training Set (80%):** Utilized for the iterative learning process and weight optimization.
- **Validation Set (10%):** Employed for monitoring generalization and hyperparameter tuning during the training loop.
- **Test Set (10%):** Reserved as a completely unseen holdout for the final, unbiased performance evaluation.

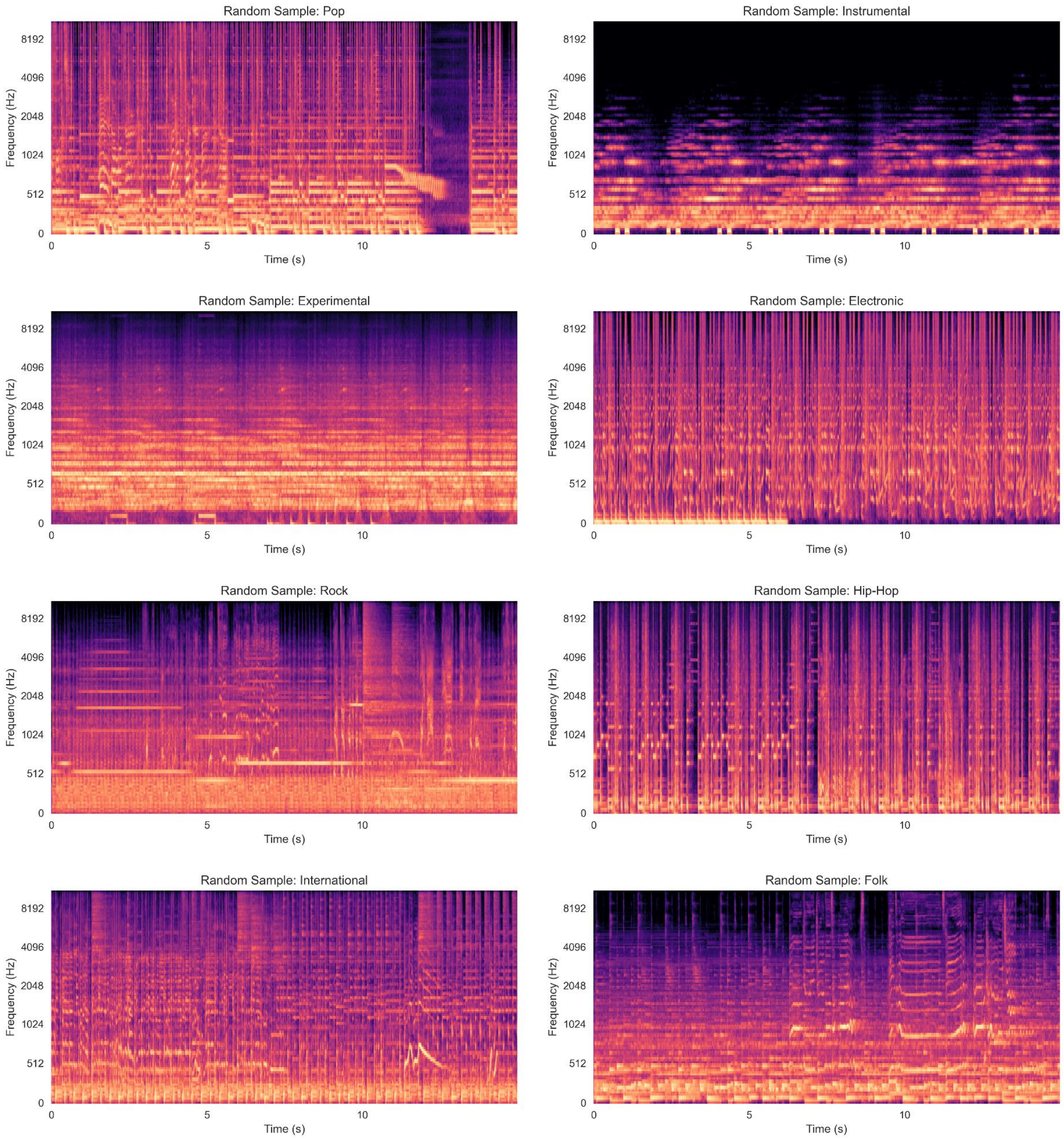
By partitioning the remaining 20% of the dataset equally (test_size=0.5), we achieve a clean 80/10/10 distribution. Furthermore, all splitting processes utilize the `random_state=42` parameter to ensure deterministic reproducibility, preventing data leakage and ensuring that reported metrics reflect the models' real-world predictive capability.

3.4. Feature extraction

For the baseline models (Random Forest, SVM, MLP), we employed a feature engineering pipeline that transforms raw audio into a vector of statistical descriptors. These include:

- **Timbral Features:** Mel-Frequency Cepstral Coefficients (MFCCs) to capture the spectral envelope.
- **Harmonic Features:** Chroma features, representing the distribution of energy across the 12 pitch classes.
- **Rhythmic/Spectral Features:** Spectral contrast and other temporal descriptors to quantify rhythmic patterns and energy dynamics.

These features were aggregated into a scaled input matrix (`X_train_ml_scaled`), forming the basis for the classical baseline evaluations. For the state-of-the-art CNN architecture, the raw audio is converted into time-frequency representations (spectrograms), allowing the network to operate directly on the spectral distribution without the loss of information inherent in manual descriptor aggregation.



Random sample spectrograms feature representation (one for each genre)

3.5. Evaluation metrics

Key Diagnostic Tools used:

- Confusion Matrix: A heatmap representing the model's predictions against actual labels. It is essential for visualizing classification bias. It indicates a high degree of feature similarity between specific categories.
- Classification Report (Precision, Recall, F1-Score):
    - Precision: how reliable the model is when it predicts a specific genre.
    - Recall: model's ability to find all instances of a genre.
    - F1-Score: harmonic mean of the two, representing the overall balance for each class.

4. Evaluation and results

4.1. Diagnostic Framework

This section provides a systematic diagnostic analysis of the classification performance. By leveraging the **Confusion Matrix**, which visualizes inter-class biases, alongside **Precision, Recall, and F1-score** metrics, we identify specific clusters of genre confusion. This evaluation distinguishes between intrinsic dataset ambiguities and feature-space limitations, establishing a clear roadmap for future refinements, such as targeted data augmentation or the integration of secondary acoustic descriptors.

4.2. Optimization and Pipeline Rigor

- **Data Preprocessing:** All features and spectrograms are normalized using a `StandardScaler` fitted exclusively on the training set to prevent **data leakage**.
- **Training Strategy:** The CNN model utilizes an `EarlyStopping` callback (patience=5) and a `ReduceLROnPlateau` scheduler (factor=0.2) to ensure optimal convergence, transitioning from aggressive initial learning to fine-tuned optimization.
- **Evaluation:** Both approaches are evaluated using stratified metrics, with performance analyzed via confusion matrices to assess genre-specific misclassifications.

4.3. Comparative Analysis

To contextualize the performance of the proposed architecture, the following list summarizes the test set accuracy across all implemented models. This analysis provides an empirical overview of the performance trajectory, shifting from classical, manually-engineered baselines to the autonomous feature learning capabilities of the state-of-the-art CNN.

- **Random Forest**:
  - Train accuracy: ~0.99
  - Test accuracy: ~0.51
  - Overfitting: High

	precision	recall	f1-score	support
Electronic	0.48	0.50	0.49	100
Experimental	0.43	0.32	0.37	100
Folk	0.55	0.66	0.60	100
Hip-Hop	0.57	0.67	0.61	100
Instrumental	0.61	0.61	0.61	100
International	0.56	0.44	0.49	100
Pop	0.33	0.27	0.30	100
Rock	0.50	0.62	0.55	100
accuracy			0.51	800
macro avg	0.50	0.51	0.50	800
weighted avg	0.50	0.51	0.50	800

- **SVM**:
  - Train accuracy: ~0.63
  - Test accuracy: ~0.52
  - Overfitting: Low

	precision	recall	f1-score	support
Electronic	0.45	0.54	0.49	100
Experimental	0.35	0.25	0.29	100
Folk	0.58	0.69	0.63	100
Hip-Hop	0.61	0.69	0.65	100
Instrumental	0.60	0.52	0.56	100
International	0.65	0.57	0.61	100
Pop	0.34	0.28	0.31	100
Rock	0.50	0.60	0.55	100
accuracy			0.52	800
macro avg	0.51	0.52	0.51	800
weighted avg	0.51	0.52	0.51	800

- **MLP**:
    - Train accuracy: ~0.82
    - Test accuracy: ~0.49
    - Overfitting: Moderate

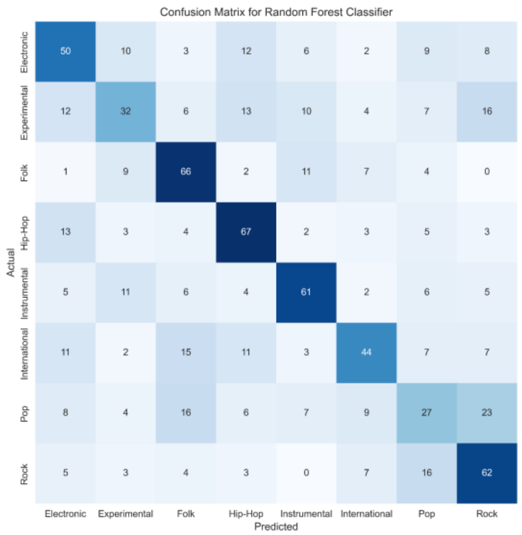
- **CNN**:
    - Train accuracy: ~0.69
    - Test accuracy: ~0.62
    - Overfitting: Low

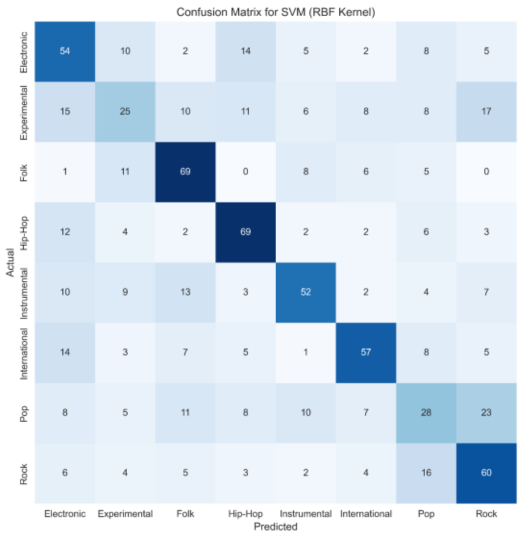
MLP:

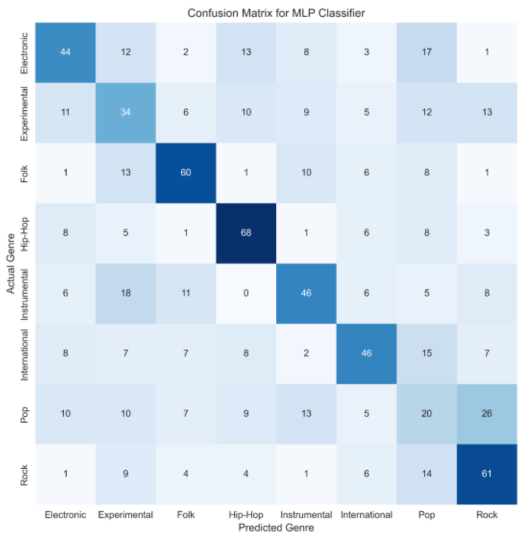
	precision	recall	f1-score	support
Electronic	0.56	0.51	0.53	100
Experimental	0.36	0.42	0.39	100
Folk	0.52	0.53	0.52	100
Hip-Hop	0.64	0.66	0.65	100
Instrumental	0.47	0.46	0.46	100
International	0.47	0.46	0.47	100
Pop	0.32	0.29	0.31	100
Rock	0.55	0.57	0.56	100
accuracy			0.49	800
macro avg	0.49	0.49	0.49	800
weighted avg	0.49	0.49	0.49	800

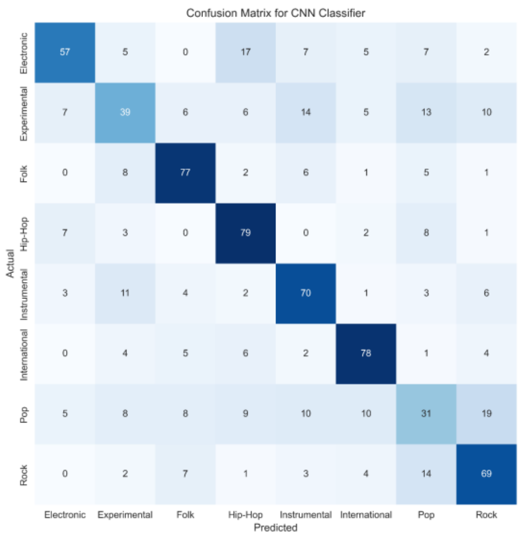
CNN:

	precision	recall	f1-score	support
Electronic	0.72	0.57	0.64	100
Experimental	0.49	0.39	0.43	100
Folk	0.72	0.77	0.74	100
Hip-Hop	0.65	0.79	0.71	100
Instrumental	0.62	0.70	0.66	100
International	0.74	0.78	0.76	100
Pop	0.38	0.31	0.34	100
Rock	0.62	0.69	0.65	100
accuracy			0.62	800
macro avg	0.62	0.62	0.62	800
weighted avg	0.62	0.62	0.62	800

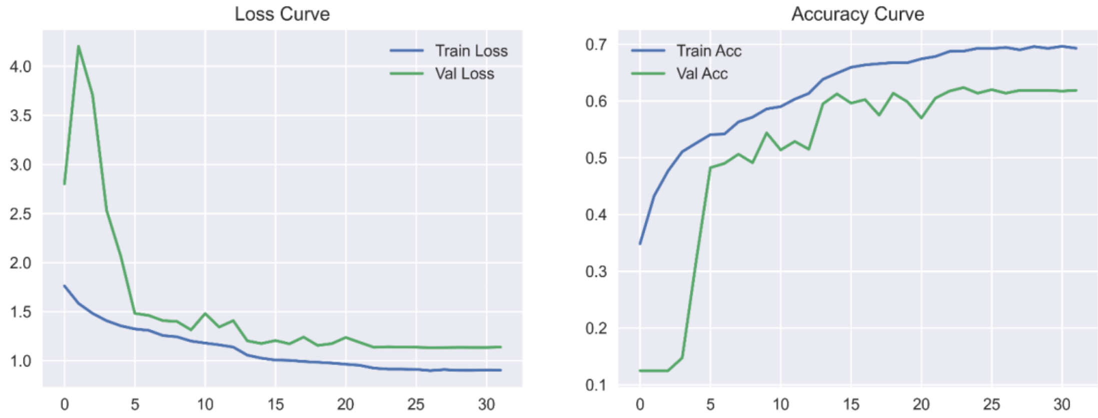








Comparison between trained models performances through confusion matrices



CNN training curves graphs

4.4. Key findings and interpretations

- **The Feature Engineering Ceiling:** The baseline Random Forest, alongside intermediate models (SVM, MLP), fluctuates within a narrow accuracy band (49–52%). Despite using mathematically distinct approaches, none could break this performance plateau. This provides strong empirical evidence that the limitation is not algorithmic, but rather intrinsic to the input data: static statistical descriptors (MFCCs, Chroma) fail to capture the hierarchical temporal and timbral nuances necessary for robust multi-genre discrimination.
- **The CNN Paradigm Shift (SOTA):** The CNN achieves a significant performance jump, yielding an accuracy of ~62%. This confirms the hypothesis that hierarchical feature extraction, where the model autonomously identifies low-level spectral textures and progresses toward high-level rhythmic and harmonic patterns, is fundamentally superior for music audio analysis compared to static statistical representations.
- **Generalization vs. Memorization:** The disparity in training dynamics is stark. While the Random Forest baseline exhibited severe overfitting (near-perfect training accuracy with poor generalization) and the MLP showed moderate overfitting, the CNN demonstrated a remarkably stable learning curve. The proximity between training and testing metrics underscores the CNN's robustness in identifying abstract, generalizable patterns rather than simply memorizing training instances.

4.5. Concluding remarks

The systematic evaluation confirms that moving from **Feature-Based Classification** (RF, SVM, MLP) to **End-to-End Deep Learning** (CNN) addresses the primary bottleneck in the music classification pipeline. By replacing manually defined features with an autonomous feature learning architecture, the model not only attains higher classification accuracy but also demonstrates a superior capacity to generalize across diverse musical genres.

This comparative study successfully justifies the adoption of convolutional architectures as the optimal SOTA solution for complex, multi-class Music Information Retrieval (MIR) tasks. The results validate the research hypothesis: the path to higher performance in genre classification lies not in the complexity of the classifier, but in the autonomous learning of hierarchical spectral features directly from the time-frequency domain.

5. Future perspectives and potential improvements

While the proposed CNN architecture demonstrates superior performance compared to classical baselines, the complexity of music genre classification presents several avenues for further research and optimization:

- **Data Augmentation Strategies:** The current model operates on static spectrograms. Incorporating time-stretching, pitch-shifting, or adding background noise during training could improve the model's invariance to recording conditions, leading to more robust generalization.
- **Architecture Expansion (Residual Learning):** Implementing ResNet-style skip connections could mitigate vanishing gradient issues, allowing for deeper networks that can capture even finer-grained harmonic textures without increasing the risk of overfitting.
- **Temporal Context Modeling:** Current 2D-CNN architectures treat the spectrogram as a static image. Integrating Recurrent Neural Networks (RNNs) or Transformers, specifically via CRNN (Convolutional Recurrent Neural Networks) architectures, could allow the model to better capture the long-term temporal dependencies and evolution of musical structure over time.
- **Attention Mechanisms:** Incorporating attention blocks would enable the model to dynamically weight the importance of specific frequency bands or time intervals, potentially improving performance on genres that are defined by transient percussive events rather than steady-state harmonic content.
- **Transfer Learning:** Utilizing pre-trained audio models and fine-tuning them on the FMA dataset could provide a significant boost in performance, especially if the available training data is limited.